General discussion of $\chi(\vec{q},\omega)$ in interacting system:

<u>RPA</u> $$\chi(\vec{q},\omega) = \frac{\chi_0(\vec{q},\omega)}{1 \pm U\chi_0(\vec{q},\omega)} \quad \begin{cases} \text{"}-\text{" for charge} \\ \text{"}+\text{" for spin} \end{cases}$$

Energy absorption is determined by $\mathrm{Im}\,\chi = \chi''$.

<u>charge</u> $$\chi' + i\chi'' = \frac{\chi_0' + i\chi_0''}{1 - U(\chi_0' + i\chi_0'')} = \frac{(\chi_0' + i\chi_0'')(1 - U\chi_0' + iU\chi_0'')}{(1-U\chi_0')^2 + (U\chi_0'')^2}$$

$$= \frac{\chi_0'(1-U\chi_0') - \chi_0''\cdot U\chi_0''}{(1-U\chi_0')^2 + (U\chi_0'')^2}$$

$$+ i\chi_0'' \frac{1 - U\chi_0' + U\chi_0'}{(1-U\chi_0')^2 + (U\chi_0'')^2}$$

$$\Rightarrow \chi'' = \frac{\chi_0''}{(1-U\chi_0')^2 + (U\chi_0'')^2}$$

Thus, absorption = physical excitation, is present if

(1) $\chi_0''(\vec{q},\omega) \neq 0$: particle-hole continuum

(2) $\chi_0''(\vec{q},\omega) = +0 \Rightarrow \chi''(\vec{q},\omega) = \pi\,\delta\left(1 - U\chi_0'(\vec{q},\omega)\right)$

Thus $1 = U\chi_0'(\vec{q},\omega)$ describes new, <u>collective</u> excitation of the system (property of interacting system only!)

Short-range interaction $U\hat{n}_\uparrow\hat{n}_\downarrow$

<u>Charge</u> susceptibility denominator: $1 - U\chi_0(\vec{q},\omega) = 0$

<u>Spin</u> $\leftarrow / \text{—} / \text{—} / \text{—} /$: $1 + U\chi_0(\vec{q},\omega) = 0$

For small $\vec{q}$ and ω enter via $s = \frac{\omega}{v_F q}$

$$\chi_0'(\vec{q},\omega) \cong 2\nu\left(-1 + \frac{1}{2} s \ln\left|\frac{1+s}{1-s}\right|\right)$$

$$\chi_0''(\vec{q},\omega) \cong \begin{cases} 0, & s > 1 \\ -\pi(2\nu), & s < 1 \end{cases}$$

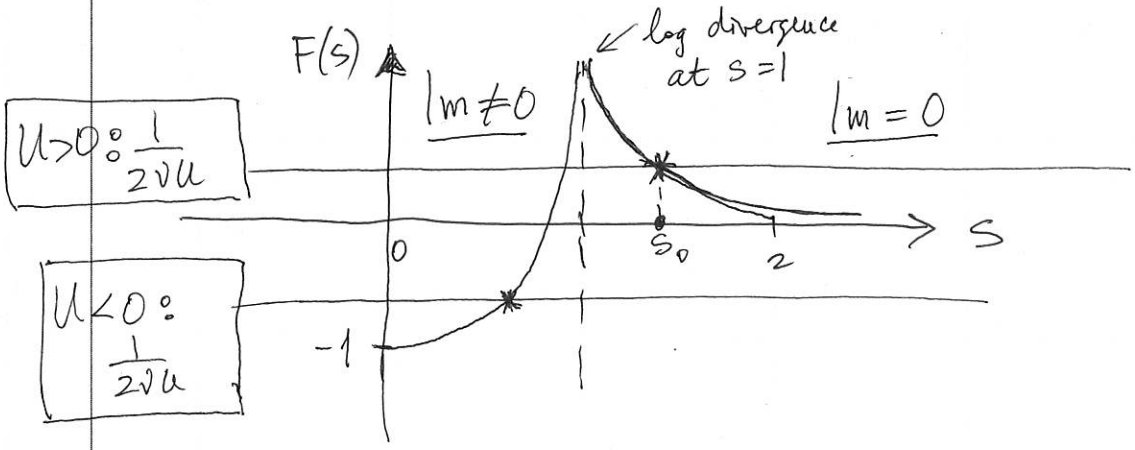
Assume linear dispersion of collective mode

i.e. $\omega = cq \Rightarrow s = \frac{cq}{v_F q} = \frac{c}{v_F}$

<u>Charge</u>: $1 = 2\nu U\left(-1 + \frac{s}{2}\ln\left|\frac{s+1}{s-1}\right| - i\pi\theta(1-s)\right)$

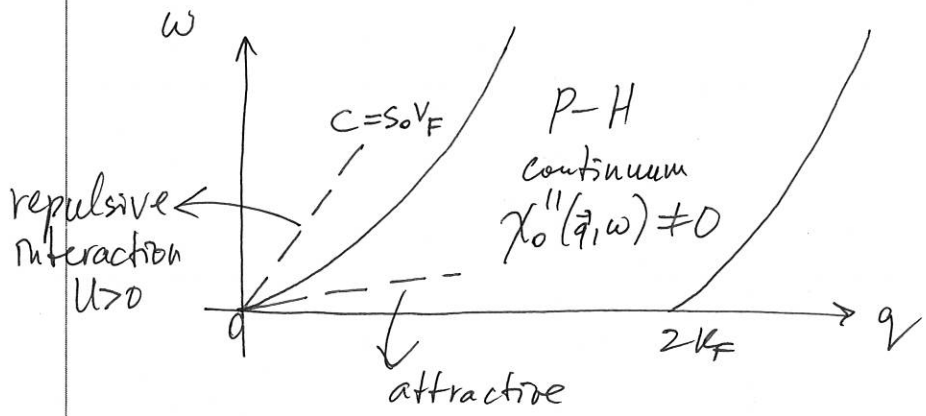
Thus for $s = \frac{c}{v_F} > 1$ the imaginary part is absent, $\theta = 0$, and there is a solution $s_0 > 1$.

$$\frac{1}{2\nu U} = -1 + \frac{s}{2}\ln\left|\frac{s+1}{s-1}\right| \equiv F(s)$$



This collective excitation is called "zero sound"

Charge sector



: zero-sound is inside P-H continuum, where $\chi_0'' \neq 0$,
$\Rightarrow$ this mode is <u>damped</u>.

<u>$U>0$:</u> $s_0 > 1$ (for any $U>0$ there is a solution!)

$\Rightarrow c = s_0 v_F > v_F$

Zero-sound is <u>above</u> P-H continuum

Describes stable and long-living collective excitation of electron density.

* Spin sector and Stoner Ferromagnetism

Previously found: transverse susceptibility of interacting Fermi gas with contact (Hubbard) interaction

$$\Pi_{\uparrow\downarrow}(q,\omega_n) = \frac{\Pi^{(0)}_{\uparrow\downarrow}(q,\omega_n)}{1+U\,\Pi^{(0)}_{\uparrow\downarrow}(q,\omega_n)}$$

where $\Pi^{(0)}_{\uparrow\downarrow}(q,\omega_n) = \frac{1}{\beta V}\sum_{\vec{p},p_n} G^{(0)}_{\uparrow}(\vec{p},ip_n)\,G^{(0)}_{\downarrow}(\vec{p}+\vec{q},ip_n+i\omega_n)$

$$= \frac{1}{\beta V}\sum_{\vec{p},p_n}\frac{1}{ip_n-\xi_{\uparrow}(p)}\,\frac{1}{ip_n+i\omega_n-\xi_{\downarrow}(\vec{p}+\vec{q})} =$$

$$= \frac{1}{\beta V}\sum_{\vec{p},p_n}\left(\frac{1}{ip_n-\xi_{\uparrow}(p)} - \frac{1}{ip_n+i\omega_n-\xi_{\downarrow}(\vec{p}+\vec{q})}\right)\frac{1}{i\omega_n+\xi_{\uparrow}(p)-\xi_{\downarrow}(p+q)}$$

Then use $\frac{1}{\beta}\sum_{p_n}\frac{1}{ip_n-\varepsilon} = n_F(\varepsilon)$ so that

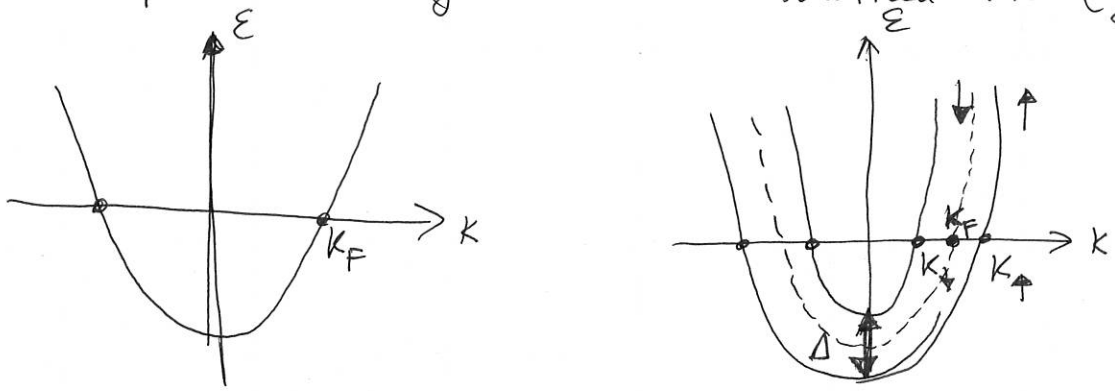
$$\Pi^{(0)}_{\uparrow\downarrow}(\vec{q},\omega_n) = \frac{1}{V}\sum_{\vec{p}}\frac{n_F(\xi_{\uparrow}(p)) - n_F(\xi_{\downarrow}(p+q))}{i\omega_n+\xi_{\uparrow}(p)-\xi_{\downarrow}(p+q)}$$

In the spin-rotation-invariant system, where $\xi_{\uparrow}(p)=\xi_{\downarrow}(p)$, $\Pi^{(0)}_{\uparrow\downarrow}$ is just a $\underbrace{\left(\tfrac{1}{2}\right)}_{\text{no summation over spin}}$ of $P^{(1)}(q,\omega_n)$ we calculated previously

But this consideration can be straightforwardly extended into spontaneously polarized <u>ferromagnetic</u> state in which density of electrons with $\uparrow$ (up) spin along the $\hat{z}$-axis is greater than the density of $\downarrow$ (down) spin electrons. That is, $n_{\uparrow} = \int\frac{d^3\vec{k}}{(2\pi)^3}\langle\psi^{+}_{\uparrow}(k)\psi_{\uparrow}(k)\rangle$

$$n_{\downarrow} = \int\frac{d^3k}{(2\pi)^3}\langle\psi^{+}_{\downarrow}(k)\psi_{\downarrow}(k)\rangle.$$

Unpolarized state ($\zeta = 0$) Polarized state ($\zeta > 0$)



To achieve this, the Fermi surfaces of ↑ and ↓ electrons must enclose different volumes, such that $n_\uparrow > n_\downarrow$ but total density $n = n_\uparrow + n_\downarrow$ fixed.

At T=0 $$n_\uparrow = \int \frac{d^3\vec{k}}{(2\pi)^3}\,(1) = \frac{4\pi}{(2\pi)^3}\int_0^{k_\uparrow} dk\, k^2 = \frac{1}{3}k_\uparrow^3\,\frac{1}{2\pi^2} = \frac{k_\uparrow^3}{6\pi^2}$$

integral over occupied states

$$n_\downarrow = \frac{k_\downarrow^3}{6\pi^2} \qquad \text{but} \quad n = \frac{2k_F^3}{6\pi^2} = \frac{k_\uparrow^3 + k_\downarrow^3}{6\pi^2}.$$

Spin polarization $$\zeta = \frac{n_\uparrow - n_\downarrow}{n} = \frac{k_\uparrow^3 - k_\downarrow^3}{k_\uparrow^3 + k_\downarrow^3} = \begin{cases} 0, & k_\uparrow = k_\downarrow \\ 1, & k_\downarrow = 0. \end{cases}$$

Thus, fully polarized gas has $\zeta = 1$: all electrons have $\sigma = \uparrow$.
In this case $2k_F^3 = k_\uparrow^3$, i.e. $k_\uparrow = 2^{1/3} k_F > k_F$.

T=0: $$n_\uparrow = \int \frac{d^3\vec{k}}{(2\pi)^2}\,\theta\Big(\mu - \big[\underbrace{\varepsilon(k)}_{=\frac{k^2}{2m}} - \tfrac{1}{2}\Delta\big]\Big) = \int \frac{d\vec{k}}{(2\pi)^3}\,\theta\Big(\mu + \tfrac{1}{2}\Delta - \frac{k^2}{2m}\Big)$$

$$= \int \frac{d\vec{k}}{(2\pi)^3}\,\theta(k_\uparrow^2 - k^2) = \frac{4\pi}{(2\pi)^3}\,\frac{1}{3}k_\uparrow^3 = \frac{k_\uparrow^3}{6\pi^2}$$

where $k_\uparrow^2 = 2m(\mu + \frac{1}{2}\Delta)$

but $\mu = \frac{k_F^2}{2m} \Rightarrow k_\uparrow^2 = k_F^2 + m\Delta$. Clearly $k_\downarrow^2 = k_F^2 - m\Delta$.
$\Rightarrow 2m\Delta = k_\uparrow^2 - k_\downarrow^2$.

P-H continuum

$$\mathrm{Im}\,\chi_0(\vec{q},\omega) = -\pi \int \frac{d\vec{k}}{(2\pi)^3} \left(f(\xi_k) - f(\omega+\xi_k)\right) \delta\left(\omega - (\xi_{k+q} - \xi_k)\right)$$

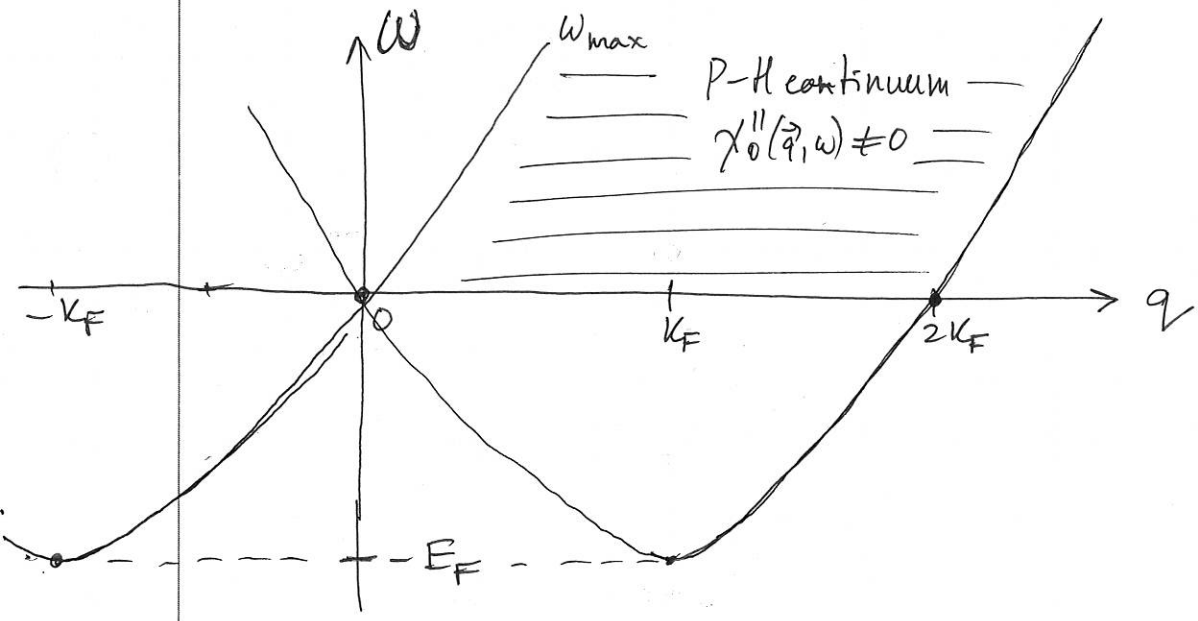
$$\omega = \xi_{k+q} - \xi_k = \frac{(\vec{k}+\vec{q})^2 - k^2}{2m} = \frac{q^2 + 2\vec{k}\cdot\vec{q}}{2m}$$

$\Rightarrow$ ω_{max} for $\vec{k}\cdot\vec{q} = k_F q$

$$\omega_{max} = \frac{q^2 + 2qk_F}{2m} = \frac{(q+k_F)^2 - k_F^2}{2m} = -E_F + \frac{(k_F+q)^2}{2m}$$

ω_{min} for $\vec{k}\cdot\vec{q} = -k_F q$

$$\omega_{min} = \frac{q^2 - 2k_F q}{2m} = \frac{(q-k_F)^2 - k_F^2}{2m} = -E_F + \frac{(k_F-q)^2}{2m}$$



Mean-field for itinerant ferromagnetism

$$U \sum_r \hat{n}_{r\uparrow} \hat{n}_{r\downarrow} = U \sum_r (\hat{n}_{r\uparrow} + n_\uparrow - n_\uparrow)(\hat{n}_{r\downarrow} + n_\downarrow - n_\uparrow)$$

operators

expectation value

$$= U \sum_r (n_\uparrow + \delta\hat{n}_{r\uparrow})(n_\downarrow + \delta\hat{n}_{r\downarrow}) = \qquad (n_\sigma \equiv \langle \hat{n}_{r\sigma} \rangle)$$

where $\delta\hat{n}_{r\sigma} \equiv \hat{n}_{r\sigma} - n_\sigma$

$$= U \sum_r \{ n_\uparrow n_\downarrow + n_\uparrow(\hat{n}_{r\downarrow} - n_\downarrow) + n_\downarrow(\hat{n}_{r\uparrow} - n_\uparrow) + \underbrace{\delta\hat{n}_{r\uparrow}\,\delta\hat{n}_{r\downarrow}} \}$$

<u>Mean-field approx:</u> neglect this quadratic in "small" fluctuations term

$$\Rightarrow U \sum_r (n_\uparrow \hat{n}_{r\downarrow} + n_\downarrow \hat{n}_{r\uparrow} - n_\uparrow n_\downarrow)$$

Re-write $\cancel{(n_\uparrow + n_\downarrow - n_\downarrow)\hat{n}_{r\downarrow} + (n_\downarrow + n_\uparrow - n_\uparrow)\hat{n}_{r\uparrow}}$

$\cancel{= (n_\uparrow - n_\downarrow)(\hat{n}_{r\downarrow} - \hat{n}_{r\uparrow})}$

$$n_\uparrow = \tfrac{1}{2} n (1+\xi), \quad n_\downarrow = \tfrac{1}{2} n (1-\xi)$$

$$\Rightarrow U \sum_r \tfrac{1}{2} n \left[(1+\xi)\hat{n}_{r\downarrow} + (1-\xi)\hat{n}_{r\uparrow} \right] - \tfrac{1}{4} n^2 (1-\xi^2)$$

$$= U \sum_r \left\{ \tfrac{1}{2} n (\hat{n}_{r\uparrow} + \hat{n}_{r\downarrow}) - \tfrac{1}{2} n \xi (\hat{n}_{r\uparrow} - \hat{n}_{r\downarrow}) - \frac{n^2(1-\xi^2)}{4} \right\}$$

$$= \underbrace{\tfrac{1}{2} n U \sum_{r\sigma} \hat{n}_{r\sigma}}_{\text{correction to chemical potential due to interaction } U.} - \underbrace{\tfrac{1}{2} n \xi U \sum_{r\sigma=\pm 1} \sigma \hat{n}_{r\sigma}}_{\text{Spin } (\sigma = \pm 1) \text{ dependent correction}} + \text{const.}$$

$n \xi U = (n_\uparrow - n_\downarrow) U \equiv \Delta$

$$\Rightarrow \hat{H}_{MF} = \sum_{k\sigma} \xi(k)\, \hat{c}^{+}_{k\sigma} \hat{c}_{k\sigma} - \underbrace{\sum_{r\sigma} \frac{\sigma}{2}\Delta\, \hat{c}^{+}_{r\sigma} \hat{c}_{r\sigma}}_{\sum_{\sigma} \frac{\sigma}{2}\Delta \sum_{k} \hat{c}^{+}_{k\sigma} \hat{c}_{k\sigma}} + \sum_{r} \frac{U}{4} n^2 \xi^2 - \frac{U}{4} n^2$$

$$= \sum_{k\sigma} \left(\xi(k) - \frac{1}{2}\sigma\Delta\right) \hat{c}^{+}_{k\sigma} \hat{c}_{k\sigma} + \underbrace{\sum_{r} \left(\frac{\Delta^2}{4U} - \frac{1}{4} U n^2\right)}_{\text{correction to the ground state energy due to finite } \Delta.}.$$

Thus, $\sum_{k} \left(\xi(k) - \frac{1}{2}\Delta\right) \hat{c}^{+}_{k\uparrow} \hat{c}_{k\uparrow}$

$+ \sum_{k} \left(\xi(k) + \frac{1}{2}\Delta\right) \hat{c}^{+}_{k\downarrow} \hat{c}_{k\downarrow}$

$\Rightarrow$ electrons with $\uparrow$ ($\downarrow$) spin ($\sigma = +1$ (-1)) have their energy shifted by $-\frac{1}{2}\Delta$ ($+\frac{1}{2}\Delta$), correspondingly.

Energy shift $\Delta \neq 0 \Rightarrow n_\uparrow - n_\downarrow \neq 0$

thus $\Delta = U(n_\uparrow - n_\downarrow)$ is proportional to the order parameter

$n_\uparrow - n_\downarrow = 2M$

$M =$ magnetization.

Dispersion: $\xi_\sigma(k) = \xi(k) - \frac{\sigma}{2}\Delta = \varepsilon(k) - \frac{\sigma\Delta}{2} - \mu$

$= \frac{\vec{k}^2}{2m} - \frac{\sigma\Delta}{2} - \mu$